From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

# FIITJEE JEE(Advanced)-2013

## ANSWERS, HINTS & SOLUTIONS
## FULL TEST – II
## (PAPER-1)

| ANSWERS KEY | | | |
|---|---|---|---|
| | PHYSICS | CHEMISTRY | MATHEMATICS |
| Q. No. | ANSWER | ANSWER | ANSWER |
| 1. | B | C | C |
| 2. | B | B | A |
| 3. | B | D | B |
| 4. | B | C | A |
| 5. | A | B | B |
| 6. | A | A | D |
| 7. | D | C | D |
| 8. | C | C | A |
| 9. | A, B | A, B | A, C |
| 10. | B, D | A, C | A, B, C |
| 11. | B, C, D | A, B, C, D | B, C |
| 12. | A, D | A, D | B, C |
| 13. | A, B, C, D | A, D | A, B, C |
| 14. | A | B | D |
| 15. | B | A | A |
| 16. | B | B | A |
| 17. | C | C | B |
| 18. | C | B | C |
| 1. | 5 | 2 | 7 |
| 2. | 4 | 4 | 4 |
| 3. | 4 | 7 | 4 |
| 4. | 2 | 3 | 0 |
| 5. | 2 | 6 | 7 |
| 6. | 8 | 8 | 4 |
| 7. | 4 | 1 | 6 |
| 8. | 3 | 2 | 2 |
| 9. | 4 | 4 | 9 |
| 10. | 1 | 4 | 1 |

# *Physics* PART – I

## SECTION – A

1. $\varepsilon = L\frac{di}{dt}$

$400 = 1 \times \frac{2}{dt}$

$\Rightarrow$ $dt = 5 \times 10^{-3}$ sec.

$Q = idt = 2 \times 5 \times 10^{-3}$

$\therefore$ $Q = 10^{-2}$

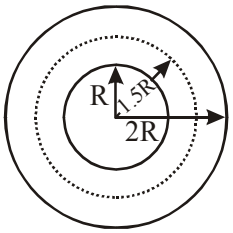
But $Q = CV$

$10^{-2} = C \times 400$

$\therefore$ $C = 25\mu F$.

2. Total flux through this area is

$= E.4\pi (1.5R)^2$




$$\frac{q_{net}}{\varepsilon_0} = \frac{1}{\varepsilon_0}\left(Q + \frac{Q\left\{(1.5R)^2 - R^3\right\}}{\left\{(2R)^3 - R^3\right\}}\right)$$

$$= \frac{Q}{\varepsilon_0}\left(1 + \frac{19}{56}\right) = \frac{75Q}{56\varepsilon_0}$$

Gauss law $E .4 \pi (1.5R)^2 = Q (75/56\varepsilon_0)$

$E = \frac{75Q}{504\pi\varepsilon_0 R^2}$

3. $w_0 = \frac{1}{\sqrt{LC}} = \frac{1}{2\times10^{-2}}$

op, $2\pi f = 50$

op, $f = \frac{25}{\pi}$Hz.

4. Use the definition of an elastic collision.

5. Buoyant force = Weight of liquid displaced = $\frac{4}{3}\pi r^2 \rho g$

Viscous force = Stoke's drag force = $6\pi\eta rv$

$\therefore$ $6\pi\eta rv = \frac{4}{3}\pi r^2 \rho g$

6. $2(k2x_0) = mg + iBL$

$mg = 2kx_0$

$i = \frac{\varepsilon}{R}$

$\therefore$ $B = \frac{mgR}{\varepsilon L}$.

8. For a internal point

$g' = \frac{gr}{R}$

$\Rightarrow g' \propto r$

F as a external point

$$g' = \frac{GM}{r^2} = g' \propto \frac{1}{r^2}$$

9. $$\frac{dx}{dt} = t^2 \qquad \ldots(i)$$

$$y = \frac{1}{2}\frac{t^3}{3}$$

$$\frac{dy}{dt} = \frac{t^2}{2} \qquad \ldots(ii)$$

$t = 1,\ v_x = 1,\ v_y = \frac{1}{2}$

$$\vec{v} = \hat{i} + \frac{1}{2}\hat{j}$$

$$\frac{d^2x}{dt^2} = 2t \qquad \ldots(iii)$$

$$\frac{d^2y}{dt^2} = t \qquad \ldots(iv)$$

at $t = 1$ s $a_x = 2$ and $a_y = 1$

$\therefore\ \vec{a} = 2\hat{i} + \hat{j}$

10. Acceleration of M, $a = \left(\frac{F}{M}\right)$

$$\ell = \frac{1}{2}\frac{F}{M}.t^2$$

$$t = \sqrt{\frac{2M\ell}{F}}\ .$$

11. $\vec{\tau} = \vec{M} \times \vec{B}$ and $U = -\vec{M}.\vec{B}$

here $\vec{M}$ and $\vec{B}$ are anti–parallel

$\therefore\ \vec{\tau} = \vec{O}$ and U = +MB (maximum)

13. $v + u = D$

and $v - u = X$

$$\Rightarrow\ v = \frac{D+X}{2}, u = \frac{D-X}{2}$$

and $$f = \frac{D^2 - X^2}{4D}$$

$$m_1 = \frac{D+X}{D-X},\ m_2 = \frac{D-X}{D+X}$$

14. - 15. The impedance is

$z = \sqrt{R^2 + \omega^2 L^2}$ and the rms

current is $i_{rms} = \frac{V_{rms}}{z}$.

16. Since the center of mass will remain at rest.

$m(5\ell - x) = M.x$

$x = \dfrac{5m\ell}{(M+m)}$.

17. $Mv_M - mv_m = 0$ …(i)

$\dfrac{1}{2}K\ell^2 = \dfrac{1}{2}Mv_M^2 + \dfrac{1}{2}mv_m^2$ …(ii)

Solving (i) and (ii)

$v_M = \sqrt{\dfrac{Km}{M(M+m)}}\ell$.

18. $v_m = \dfrac{M}{m}\sqrt{\dfrac{Km}{M(M+m)}}\ell$

$= \sqrt{\dfrac{KM}{m(M+m)}}\ell$

$KE = \dfrac{1}{2}mv_m^2$

## SECTION – C

1. Let, $V_0 \rightarrow$ velocity of sound

when source is O at the velocity component of source towards listener is $V_S \cos 90 = 0$

$\therefore \quad F' = F_0\left(\dfrac{V_0}{V_0 - V_S\cos 90}\right) = F_S$

time taken by sound to reach listener $= \dfrac{\ell}{V_0}$.

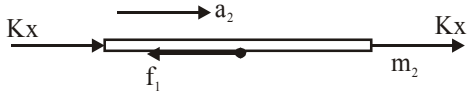
Distance moved by source in this time

$= \left(\dfrac{\ell}{V_0}\right)V_S$

$\therefore$ Distance between source and listener

$= \sqrt{\ell^2 + n^2\ell^2} = \ell\sqrt{1+n^2}$

$= 105$ m. 2. If the plank is displaced slightly by x towards left, then



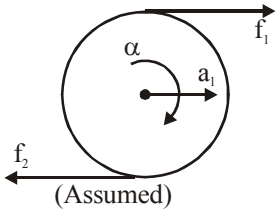

$2Kx - f_1 = m\,a_2$ …(i)

$f_1 - f_2 = 8\,m\,a_1$ …(ii)

$\tau_{cm} = I_{cm}\,\alpha$

or $(f_1 + f_2)R = \dfrac{(8m)R^2}{2}\alpha$

$(f_1 + f_2) = \dfrac{(8m)a_1}{2}$ …(iii)




From (ii) and (iii),

$2f_1 = (8m)\times\dfrac{3a_1}{2}$

or $f_1 = (8m)\times\dfrac{3a_1}{4}$

and $f_2 = 2\,ma_1$.
Hence, $f_2$ will be in forward direction.
Also, for no slipping, at the point of contact between plank and cylinder.

$2a_1 = a_2$ …(iv)

$$f_1 = (8m)\times\left(\frac{3a_2}{8}\right) = (3m)a_2 \quad \text{…(v)}$$

from (i) and (v),

$2Kx - (3m)\,a_2 = m\,a_2$

$$a_2 = \frac{2K}{4m}x$$

or $$\left(\frac{-d^2x}{dt^2}\right) = \left(\frac{2K}{4m}\right)x$$

or $$\frac{d^2x}{dt^2} = -\left(\frac{2}{4}\times\frac{\pi^2}{2}\right)x$$

or $$\omega^2 = \frac{\pi^2}{4}$$

or $$\omega = \frac{\pi}{2}$$

or $$\frac{2\pi}{T} = \frac{\pi}{2}$$

or $T = 4$ seconds.

3. When the two disc come in contact, they exert equal and opposite forces.

$$\int F\,dt = I_1(\omega_0 - \omega_1) \quad \text{…(i)}$$

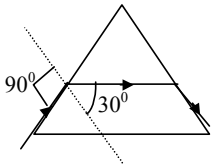
$$\int F\,b\,dt = I_2\omega_2 \quad \text{…(ii)}$$

finally $\omega_1 a = \omega_2 b$

$$I_1(\omega_0 - \omega_1) = I_2\frac{a^2}{b^2}.\omega_1$$

$$\omega_1 = \frac{I_1\omega_0}{I_1 + \frac{a^2}{b^2}I_2} = 4\ \text{rad/s}.$$

4 $$\mu = \frac{\sin 90^0}{\sin 30^0} = 2$$




5. Max$^n$ energy is liberated for transition

$E_n \to E_1$

and minimum energy for $E_n \to E_{n-1}$

Hence,

$$\frac{E_1}{n^2} - \frac{E_1}{12} = 52.224\,\text{eV}$$

and $$\frac{E_1}{n^2} - \frac{E_1}{(n-1)^2} = 1.224\ \text{eV}$$

Solving we get,

$E_1 = -54.4$ eV

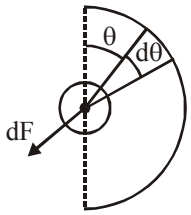
and $n = 5$

hence,

$$E_1 = -\frac{13.6Z^2}{12} = -54.4$$

$Z = 2$.

6. Magnetic field due to small ring at distance R from the centre $B = \frac{\mu_0}{4\pi}\frac{M}{R^3}$




where $M = i\,\pi\,a^2$

$$\therefore \quad B = \frac{\mu_0}{4\pi}\frac{i\pi a^2}{R^3} = \frac{\mu_0 i a^2}{4R^3}$$

$$dF = i_0 R d\theta \frac{\mu_0 i a^2}{4R^3}$$

$dF_x = dF\sin\theta$

$$= \frac{\mu_0 i . i_0 a^2 \sin\theta d\theta}{4R^2}$$

$$F_x = \frac{\mu_0 i\, i_0 a^2}{4R^2}\int_0^{\pi} \sin\theta\, d\theta$$

$$F_x = \frac{\mu_0 i\, i_0 a^2}{4R^2} \times 2$$

$$F_x = \frac{\mu_0 i\, i_0 a^2}{2R^2}$$

$F_y = 0$

$$\therefore \quad F_{net} = F_x = \frac{\mu_0 i\, i_0 a^2}{2R^2}$$

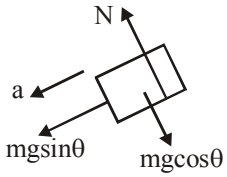
= 8 Newton.

7. F.B.D of m

$N = mg\cos\theta$ ….(i)

$mg\sin\theta = ma$

$a = g\sin\theta$




F.B.D. of M

$R = N\cos\theta + Mg$

$\mu R = N\sin\theta$

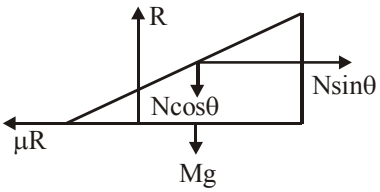
$\mu . (N\cos\theta + Mg) = N\sin\theta$

$\mu Mg = N[\sin\theta - \mu\cos\theta]$

$\mu Mg = mg\cos\theta[\sin\theta - \mu\cos\theta]$

$$m = \frac{\mu M}{\cos\theta[\sin\theta - \mu\cos\theta]}$$

$m = 4$ kg.




8. Thermal resistance of the rod,

$$R = \frac{\ell}{kA}$$

When heat is transferred from first vessel to second, temperature of first vessel decreases while that of second vessel increases. Due to both there reasons, difference between temperature of vessels decreases.

Let at an instant t, the temperature difference between two vessels be $\theta$.

$$H = \frac{\theta}{R} = \frac{KA\theta}{\ell}$$

$$dQ = Hdt = \frac{KA\theta}{\ell}dt \qquad \ldots(i)$$

Since gases are contained in two vessels, therefore, processes on gases in two vessels are isochoric.

Hence, decrease in temperature of gas in first vessel,

$$\Delta\theta_1 = \frac{dQ}{nC_V} = \frac{dQ}{2\times\frac{5R}{2}} = \frac{dQ}{5R}$$

Increase in temperature of gas in second vessel is

$$\Delta\theta_2 = \frac{dQ}{4\times\frac{3R}{2}} = \frac{dQ}{6R}$$

$\therefore$ Decrease in temperature difference $(-d\theta) = \Delta\theta_1 + \Delta\theta_2$

$$-d\theta = \frac{dQ}{R}\times\frac{11}{30}$$

or, $$-d\theta = \frac{KA\theta\times 11}{30R\ell}dt$$

or $$-\int_{50}^{25}\frac{d\theta}{\theta} = \frac{KA\times 11}{30\ell R}\int_0^t dt$$

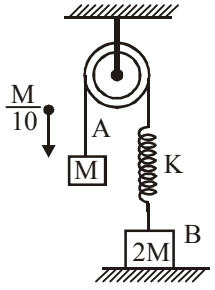
or, $$\ell n2 = \frac{KA\times 11}{30\ell R}t$$

or $$t = \frac{.693\times 30\times 242\times 10^{-2}\times 8.3\times 7}{693\times 7\times 22\times 8.3\times 10^{-4}\times 11} = 3 \text{ seconds.}$$

9. $$V_L = \left|\left(\frac{di_R}{dt} - \frac{d^2q}{dt^2}\right)\right| = L\left|\frac{d}{dt}i_R - \left(C\frac{d^2V}{dt^2} + \frac{2dC}{dt}\frac{dV}{dt}\right)\right| = 4\ \mu V.$$

10. From C.L.M.

$$\frac{M}{10}v_0 = \frac{11}{10}Mv \Rightarrow v = \frac{v_0}{11} \qquad \ldots(i)$$




For block B to leave ground

$$K(x_0 + x) = 2\ Mg \text{ (where } x_0 = \frac{Mg}{K})$$

$$\therefore\ x = \frac{Mg}{K} \qquad \ldots(ii)$$

From COE

$$\frac{11}{10}Mgx - \left[\frac{1}{2}K(x_0+x)^2 - \frac{1}{2}Kx_0^2\right] = 0 - \frac{1}{2}\frac{11}{10}Mv^2$$

solving $v_0 = g\sqrt{\frac{88M}{K}}$

putting the value

$v_0 = 1\ ms^{-1}$.

# Chemistry PART – II

## SECTION – A

1. $E_{cell} = 0 - \frac{0.0591}{1}\log\frac{1}{2}$

After electrolysis $E_{cell} = 0 - \frac{0.0591}{1}\log\frac{(1-0.5)}{(2-1)}$

No change.

2. $\sqrt{\frac{8RT}{\pi M_A}} = \sqrt{\frac{3RT}{M_B}} \Rightarrow \frac{8}{\pi M_A} = \frac{3}{M_B}$

10. $H^+ + e^- \longrightarrow \frac{1}{2}H_2 \qquad E^\circ = 0.0V$

$E = E_{H^+/H_2} = -\frac{0.059}{1}\log\frac{1}{[H^+]}$

= - 0.059 × 7 = - 0.413

So, $E_{H^+/H_2} < E_{Ni^{2+}/Ni}$

Ni will deposit at cathode

At anode cell reaction

$4OH^- \longrightarrow 2H_2O + O_2 + 4e$

11. Both complexes have $Cr^{+3}(d^3)$, and coordination number is 6. Hence $d^2sp^3$

$Pd^{+2}(d^8)$ complexes with coordination number → 4 are square plannar and magnetic moment is zero.

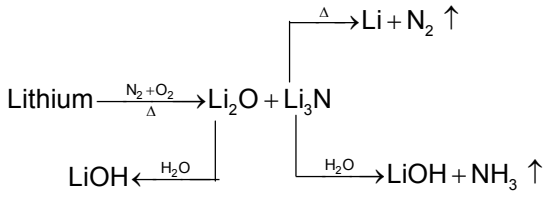
12. $FeCl_3 \xrightarrow{\Delta} FeCl_2 + Cl_2$

Dilute $CuCl_2$ is blue due to $[Cu(H_2O)_4]^{2+}$. $PH_3$ is obtained be heating white P with NaOH.

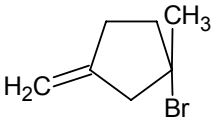
All are reducing because of weak P – H bond.

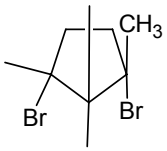
15. Bottles 1,2,3,4 have $Pb(NO_3)_2$, HCl, $Na_2CO_3$ and $CuSO_4$
When mixing
Bottle 1 + Bottle 2 → $PbSO_4$
Bottle 1 + Bottle 3 → $PbCO_3$
Bottle 1 + Bottle 4 → $PbSO_4$
Bottle 2 + Bottle 3 → $CO_2$
Bottle 2 + Bottle 4 → No visible reaction
Bottle 3 + Bottle 4 → $CuCO_3$

18.

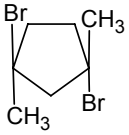


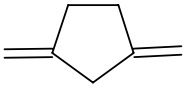
## SECTION – C

1. A = 

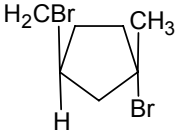

B = 


Inactive

C = 


D = 

Inactive

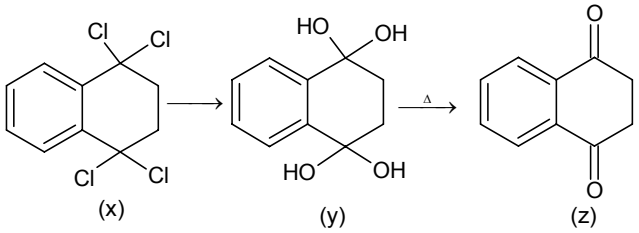
E = 


4. $\Delta T_b = K_b \times m \times i$

$2.08 = 0.52 \times 1 \times i$

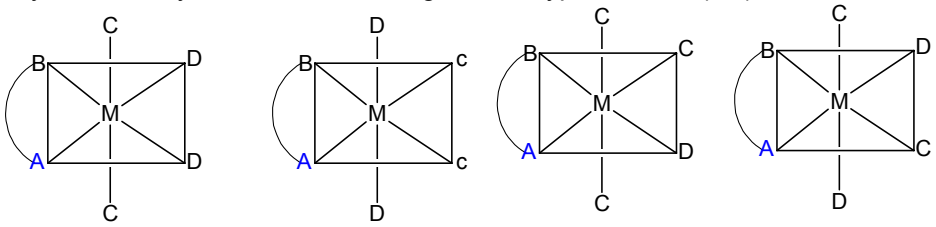
$i = 4$

$K_3\left[Fe^{III}(CN)_6\right]$

6. 


7. logK = 0.1

10. Glycinate is asymmetric bidentate ligand. AB type, so for $M(AB)C_2D_2$:



# Mathematics PART – III

## SECTION – A

1. $\left[k^2\left(\frac{\sin 3x}{3}+\frac{1}{2}\sin x\right)-\cos x-2k\sin x\right]_0^{\pi/2} \le 1$

$\left\{k^2\left(\frac{-1}{3}+\frac{1}{2}\right)-0-2k\right\}-\{-1\} \le 1$

$\frac{k^2}{6}-2k+1 \le 1$

$k(k-12) \le 0$

$0 \le k \le 12$

$k = 1, 2, 3, 4, \ldots, 12.$

2. $I_1 = \int_{-100}^{101} \frac{dx}{(5+2x-2x^2)(1+e^{2-4x})} = \int_{-100}^{101} \frac{dx}{(5+2(1-x)-2(1-x)^2)(1+e^{2-4(1-x)})}$

$= 2I_1 = \int_{-100}^{101} \frac{dx}{5+2x-2x^2} = I_2$

$\Rightarrow \frac{I_1}{I_2} = \frac{1}{2}.$

3. $I = \int \frac{2x(x-6f(x))+f(x)}{(6f(x)-x)(x^2-f(x))^2}\,dx$

$I = -\int \frac{2x}{(x^2-f(x))^2}\,dx + \int \frac{f(x)}{(6f(x)-x)(x^2-f(x))^2}\,dx = \int\left(\frac{f'(x)-2x}{(x^2-f(x))^2}\right)dx = \frac{1}{x^2-f(x)}+c$

4. Let $g(x) = (\sin x)^{\ln x} \Rightarrow g'(x) = \lim_{h\to 0}\left(\frac{(\sin(x+h))^{\ln(x+h)}-(\sin x)^{\ln x}}{h}\right)$

$\Rightarrow f(x) = g'(x) = g(x)\left(\frac{\ln(\sin x)}{x}+\cot x \ln x\right) \Rightarrow f\left(\frac{\pi}{2}\right) = g'\left(\frac{\pi}{2}\right) = 0$

5. $n(x_1) = 3$, $x_1 = 2, 3, 4.$

$4 \le x_2 < x_3 \le 8 \Rightarrow$ from 4, 5, 6, 7, 8

Selecting any 2 $\Rightarrow n(x_2, x_3) = {}^5C_2 = 10$ and $n(x_4) = 2$

$x_4 = 0, 5$

$\Rightarrow$ Number of such 4 digit numbers $3 \times 10 \times 2 = 60$

6. Let X denotes the number of toss required.

$P(X \ge 4) = \left(\frac{2}{3}\right)^3\cdot\frac{1}{3}+\left(\frac{2}{3}\right)^4\cdot\frac{1}{3}+\left(\frac{2}{3}\right)^5\cdot\frac{1}{3}+ \ldots\ldots\infty = \frac{\left(\frac{2}{3}\right)^3\cdot\frac{1}{3}}{1-\frac{2}{3}} = \frac{8}{27}$

$$P(X \geq 7) = \left(\frac{2}{3}\right)^6 \cdot \frac{1}{3} + \left(\frac{2}{3}\right)^7 \cdot \frac{1}{3} + \ ..... \infty \ = \left(\frac{\left(\frac{2}{3}\right)^6 \cdot \frac{1}{3}}{1-\frac{2}{3}}\right) = \left(\frac{2}{3}\right)^6$$

$$\text{Required probability} = \frac{\left(\frac{2}{3}\right)^6}{\left(\frac{2}{3}\right)^3} = \frac{8}{27}$$

8. $R = P'Q^8P = A^8$

$$A^2 = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2\sqrt{3}-2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3 & -2\sqrt{3}-2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} (\sqrt{3})^3 & -2(3)-2\sqrt{3}-2 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow$ If $\left[\lambda_{ij}\right]_{2\times 2} = A^k$ then $\lambda_{11} = (\sqrt{3})^k \Rightarrow r_{11} = (\sqrt{3})^8 = 81$

9. $r_1 + r_2 = 3R \Rightarrow \left(\frac{\Delta}{s-a}\right) + \left(\frac{\Delta}{s-b}\right) = 3\left(\frac{abc}{4\Delta}\right)$

$$\Rightarrow \frac{\Delta^2}{(s-a)(s-b)} = \frac{3ab}{4} \Rightarrow 4s(s-c) = 3ab$$

$$\Rightarrow \cos c = \frac{1}{2} \Rightarrow \angle c = 60^\circ$$

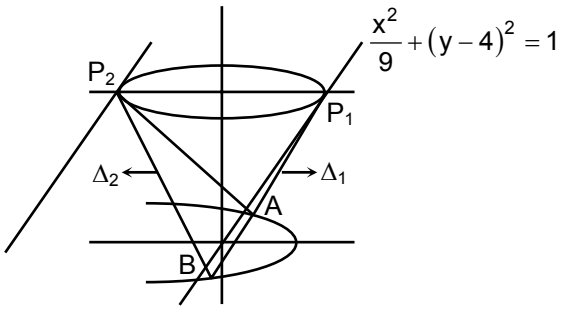
Also $r_2 + r_3 = 2R$

$$\Rightarrow \left(\frac{\Delta}{s-b}\right) + \left(\frac{\Delta}{s-c}\right) = \frac{2abc}{4\Delta}$$

$\Rightarrow 2s(s-a) = bc \Rightarrow \angle A = \frac{\pi}{2}$ and $\angle B = \frac{\pi}{6}$

10. $\Rightarrow \Delta_1 = 2\sqrt{2}\left(4-\sqrt{10}\right)$

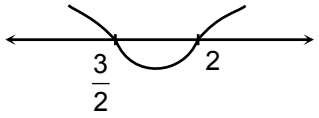
and $\Delta_2 = 2\sqrt{2}\left(4+\sqrt{10}\right)$

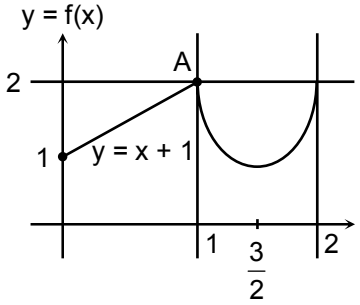



11. $(3a + 4b)(3a - 2b) = 0$ and $(3a - 2b)(a + b) = 0$

$\Rightarrow 3a = 2b$

12. f(x) is non differential at x = 1 (Point A in figure)
But f(x) is cont. $\forall\ x \in [0, 2]$
$\Rightarrow g'(t) = f(t) - f(t-1),\ t \in [1, 2]$
$= (2t^2 - 6t + 6) - ((t-1)+1)$
$= 2t^2 - 7t + 6 = (2t-3)(t-2)$




$\Rightarrow$ Maximum when $t = \frac{3}{2}$ and minimum when t = 2.

14. $P\left(\frac{A}{B_E}\right) = \frac{P(A \cap B_E)}{P(B_E)}$, where A = sum is 4.

And $P(B_E) = P(B_2) + P(B_4) + .....$ = $\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + .....\infty = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{1}{4} \cdot \frac{4}{3} = \frac{1}{3}$

$P(A \cap B_E) = P(A \cap B_2) + P(A \cap B_4) = P(B_2)P\left(\frac{A}{B_2}\right) + P(B_4)P\left(\frac{A}{B_4}\right)$

= $\frac{1}{4}\left(\frac{3}{36}\right) + \frac{1}{16}\left(\frac{1}{6^4}\right) = \frac{(4 \cdot 3 \cdot 6^2 + 1)}{16.6^4} = \frac{433}{16.6^4}$

Hence $P\left(\frac{A}{B_E}\right) = \frac{433}{16.6^4} \cdot 3 = \frac{433}{32 \cdot 216} \cong \frac{1}{16}$

15. $n(S) = 6 \cdot 6 \cdot 6 = 6^3 = 216$
Now greatest number is 4, so atleast one of the dice shows up 4.
$\therefore$ $n(A) = 4^3 - 3^3 = 37$
Hence $P(A) = \frac{37}{216}$

16-18. Equation of tangent of slope m to $y^2 = 4x$ is

$$y = mx + \frac{1}{m} \qquad \text{..... (1)}$$

16. As (1) passes through P(6, 5), so $5 = 6m + \frac{1}{m}$

$\Rightarrow 6m^2 - 5m + 1 = 0 \Rightarrow m = \frac{1}{2}$ or $m = \frac{1}{3}$

Points of contact are $\left(\frac{1}{m_1^2}, \frac{2}{m_1}\right)$ and $\left(\frac{1}{m_2^2}, \frac{2}{m_2}\right)$

Hence R(4, 4) and Q(9, 6)

Area of $\Delta PQR = \frac{1}{2}\begin{vmatrix} 6 & 5 & 1 \\ 4 & 4 & 1 \\ 9 & 6 & 1 \end{vmatrix} = \frac{1}{2}$

17. $y = \frac{1}{2}x + 2 \Rightarrow x - 2y + 4 = 0$ ..... (2)

and $y = \frac{1}{3}x + 3 \Rightarrow x - 3y + 9 = 0$

Now equation of circle $C_2$ touching $x - 3y + 9 = 0$ at (9, 6), is

$(x-9)^2 + (y-6)^2 + \lambda(x - 3y + 9) = 0$

As above circle passes through (1, 0), so

$64 + 36 + 10\lambda = 0 \Rightarrow \lambda = -10$

Circle $C_2$ is $x^2 + y^2 - 28x + 18y + 27 = 0$ ..... (3)

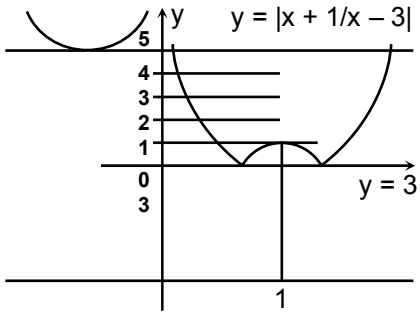
Radius of $C_2$ is $r_2^2 = 196 + 81 - 27 = 277 - 27 - 250 \Rightarrow r_2 = 5\sqrt{10}$

## SECTION – C

1. $\Rightarrow p - 3 = 0, 2, 3, 4$

$\Rightarrow \Sigma p = 5 + 6 + 7 + 3 = 21$

$\Rightarrow n = 7$




2. $\left(x + \frac{1}{\alpha^2}\right)\left(x + \frac{1}{\beta^2}\right) + \frac{1}{\alpha\beta} = x^2 + \left(\frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}\right)x + \frac{1}{(\alpha\beta)^2} + \frac{1}{\alpha\beta} = x^2 + \left(\frac{t^4 + 4t}{4t^2}\right)x + \frac{1}{4t^2} - \frac{1}{2t}$

Let $I = \int_{-1}^{2} x^2 dx + \frac{3}{2}\left(\frac{t^2}{4} + \frac{1}{t}\right) + 3\left(\frac{1}{4t^2} - \frac{1}{2t}\right)$

$\Rightarrow \frac{dI}{dt} = \frac{3}{2}\left(\frac{t}{2} - \frac{1}{t^2} - \frac{1}{t^3} + \frac{1}{t^2}\right) = 0 \Rightarrow t^4 = 2 \Rightarrow t = \pm(2)^{1/4}$

$\Rightarrow$ Min I(t) = $3 + \frac{3}{2}\left(\frac{\sqrt{2}}{4} + \frac{1}{2^{1/4}} + \frac{1}{2\sqrt{2}} - \frac{1}{2^{1/4}}\right) = 3 + \frac{3}{2}\cdot\frac{\sqrt{2}}{2} = 3 + \sqrt{\frac{9}{8}}$

$\Rightarrow \left[\frac{9}{8}\right] + 3 = 1 + 3 = 4$

3. The population after 2n days = $P\left(\left(\frac{3}{2}\cdot\frac{1}{2}\right)\cdot\left(\frac{3}{2}\cdot\frac{1}{2}\right)\cdot\text{.....n times}\right) = 3^{10}$

$\left(\frac{3}{4}\right)^n = \left(\frac{3}{4}\right)^{10} \Rightarrow n = 10 \Rightarrow$ Total number of days = 20 days

4. Let $f(x) = 2x^3 - 3x^2 + p$

$\Rightarrow f'(x) = 6x^2 - 6x = 0$ at x = 0, 1

$\Rightarrow f(0) > 0 \Rightarrow p > 0$

$f(1) < 0 \Rightarrow 2 - 3 + p < 0$

$\Rightarrow p < 1$

$\Rightarrow p \in (0, 1)$

$\Rightarrow [p] = 0$

5. Required plane is $p_1 + \lambda p_2 = 0$ and passes through (1, 2, 0)

$\Rightarrow p = \vec{r} \cdot (3\hat{i} - \hat{j} + 3\hat{k}) = 1) \Rightarrow a + b + c = 7$

6. tangent at $\left(\frac{1}{4}, 1\right)$ is $y = 2x + \frac{1}{2}$

$\Rightarrow k = \frac{19}{4}$

7. The polynomial is an everywhere differentiable function. Therefore, the points of extremum can only be roots of the derivative. Furthermore, the derivative of a polynomial is a polynomial. The polynomial of the least degree with roots $x_1 = 1$ and $x_2 = 3$ has the from

$a(x - 1)(x - 3)$.

Hence $P'(x) = a(x - 1)(x - 3) = a(x^2 - 4x + 3)$ since at the point x = 1, there must be P(1) = 6, we have

$$P(x) = \int_1^x P'(x)dx + 6 = a\int_1^x (x^2 - 4x + 3)dx + 6 = a\left(\frac{x^3}{3} - 2x^2 + 3x - \frac{4}{3}\right) + 6$$

The coefficient 'a' is determined from the condition P(3) = 2, whence a = 3.

Hence $P(x) = x^3 - 6x^2 + 9x + 2$

Now $P(2) = 8 - 24 + 18 + 2 = 28 - 24 = 4$

Also $P'(x) = 3(x^2 - 4x + 3) \Rightarrow P'(0) = 9$

$\Rightarrow P(2) + P'(0) = 4 + 9 = 13$

8. f(x) is an odd function

$\Rightarrow$ g(x) is $f^{-1}(x)$

$$\Rightarrow \text{Area} = \int_1^{2\pi} f^{-1}(y)dy = \int_0^{2\pi} f'(x)dx = \left[x(f(x))\Big|_0^{2\pi} - \int_0^{2\pi} (x + \sin x)dx\right] = 2\pi^2.$$

9. $$I_n = \left(\frac{e^{-x}(\sin x)^n}{-1}\right)_0^{\infty} + \int_0^{\infty} n(\sin x)^{n-1}\cos x e^{-x}dx$$

$$= 0 + n\left(\frac{(\sin x)^{n-1}\cos x e^{-x}}{-1}\right)_0^{\infty} + n\int_0^{\infty}\left[(\sin x)^{n-1}(-\sin x) + (\cos x)(n-1)\cos x(\sin x)^{n-2}\cos x\right]e^{-x}dx$$

$$= n\int_0^{\infty}\left(-(\sin x)^n + (n-1)(1-\sin^2 x)(\sin x)^{n-2}\right)e^{-x}dx = \frac{n(n-1)}{n^2+1}I_{n-2}$$

$$101\frac{I_{10}}{I_8} = 101 \times \frac{10 \times 9}{10^2 + 1} = 90.$$

10. $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 2(a^2 + b^2 + c^2) - 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

$= 2(4 + 9 + 16) - 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 58 - 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$.

Now $(\vec{a} + \vec{b} + \vec{c})^2 \geq 0$

$\Rightarrow a^2 + b^2 + c^2 \geq -2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

$\Rightarrow -2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) \leq 29$

$\Rightarrow |\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 \leq 87.$